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Performance assessment of commercial circadian personal exposure devices

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Through the influence on circadian rhythms, natural and artificial lighting, as well as lifestyle and architecture, affect health and performance. Epidemiological and interventional studies of light-initiated circadian biological outcomes require robust 24-hour data on personal light exposures, including blue light weighted irradiance data. The performance of the detection systems used is a key factor. We assessed the performance of 16 Actiwatch Spectrum™ devices for spectral response, directional response and dynamic range and propose techniques for calibration, deployment and data analysis for use of the watches in circadian studies. The results are presented, followed by a discussion of applicability focussing on spectral response for circadian studies.

1. Introduction

Through the influence on circadian rhythms, natural and artificial lighting, as well as lifestyle and architecture, affect health and performance.[1–4] Epidemiological and interventional studies of light-initiated circadian biological outcomes require robust 24-hour data on personal, i.e. individual light exposures and light environments, including blue light weighted irradiance data in a range of different light conditions.[5] Demonstrations of causality in these studies are limited to those aspects of the lighting environment that can be resolved from exposure measurements, so the performance of the detection systems used is a key factor. Please note that blue light is used to refer to the human circadian action spectrum throughout this paper. References to models of this action spectrum and to the blue light hazard function are explicitly made distinct in the text.

A personal exposure meter for collecting 24-hour data including irradiance data from the blue region of the visible spectrum is commercially available from the Philips Respironics Actiwatch Spectrum™ (Philips Healthcare, Best, NL). These devices, the 'spectral watches', also provide irradiance data from the red and green regions of the spectrum and activity counts from a motion counter. This paper examines the optical sensory performance by considering the spectral response characteristics, directional (i.e. angular or off-axis) response[6] and spectrally weighted irradiance response (i.e. linearity and dynamic range) of 16 spectral watches, which had been purchased for a separate research project. The specifications available did not include spectral sensitivity curves, directional response characteristics or linearity. The use of filters and diffusers to improve or modify the performance is also discussed.

Exposure measurements can be used to establish calibrated radiological measurements or as a direct proxy for exposures that may elicit photobiological responses.[7] The latter approach requires assumptions about the optical performance of the

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biological system. As a result, the limitations of the personal exposure meters are sometimes not considered. Information about the sensitivity of personal exposure meters when exposed to different intensities and spectral contents can inform these assumptions. The discussion examines the appropriateness of the optical performance of the spectral watches for circadian studies, in particular considering the spectral sensitivity of melanopsin *in vivo*.

The directional response of personal exposure meters should be considered when interpreting the data that they produce. The directional response may not be adequate for specific radiological quantities and understanding the actual contribution of off-axis light sources will require knowledge of the directional response. Moreover, biological exposures will be affected by eye movements and the selective directional response characteristics of the circadian elements of the eye, including, for example, the distribution of intrinsically photosensitive retinal ganglion cells.[8] Many of these factors will not be closely matched by personal exposure meters. Clearly, exposure patterns of a personal exposure meter may also depart from the eyes' exposure due to where the meter is worn, frequency of movement, adaptation of the eye, the eyelids as well as shading from bedding and clothing including sleeves, hats and eyewear.

As an alternative to being worn by participants in a study, personal exposure meters (and other similar equipment) can be used to measure the potential time-based exposures at selected locations within a given architectural space together with more accurate spectral spot measurements at these locations under controlled light conditions in order to interpret time-based exposures. Personal exposure meters worn by participants in a study can then be assessed relative to this information along with other known parameters of the lighting and architectural spaces.

The performance of personal exposure meters should form part of an interpretation of photobiological studies into everyday environments and personal exposures. 24-hour data on lighting in architectural spaces can aid the design of such studies and, in particular, in the selection of an appropriate intervention.[9,10]

2. Method

A 100-W collimated tungsten halogen lamp was used with a selection of neutral density filters and a diffraction grating (1200 grooves per mm with a blaze wavelength of 500 nm) to create a variety of narrow-band and broad-band illumination conditions. The spectral watches were exposed to these conditions for 90 seconds to determine the linearity and spectral sensitivity over at least five independent data points, which were logged at 15 second intervals.

All spectral measurements of the illumination conditions were made in the spectral range 360–720 nm using a Bentham DM150 scanning double grating spectroradiometer (Bentham Instruments Ltd, Reading, Berks, UK) in a controlled laboratory environment (darkened, $T = 23^{\circ}\mathrm{C}$). The spectroradiometer was calibrated using a 1000-W tungsten halogen lamp, calibrated for spectral irradiance to the National Institute of Standards and Technology (NIST) traceable reference standard.

To improve the spectral uniformity of the beam incident at the input optics of the monochromator, and to ensure a close match to the spectral content of the light incident on the sensors of the spectral watches, the input optics of the monochromator were under-filled. As only a portion of the input optics area was illuminated, the calibration underestimated the irradiance by a constant factor, due to the geometry. The illumination did not have a sharp edge, so it was not practicable to determine the under-filling factor from direct

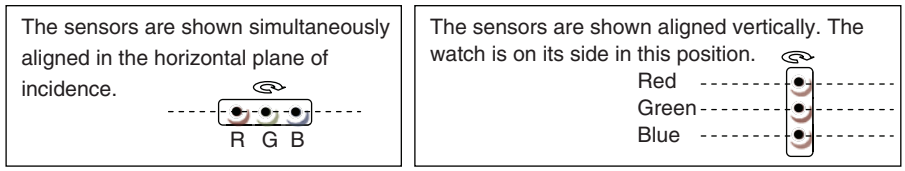


Figure 1 Orientation of the sensors for directional response for horizontal (left) and vertical (right) sensor alignments. The dotted lines represent the projection of the plane of incidence onto the face of a spectral watch. The arrows represent the axes of rotation of the sensors

measurements and it was necessary only to confirm later that the output from the spectral watches was within the linear region. The spectral measurements, thus, provided relative rather than absolute spectral irradiance data of the narrow-band illumination conditions, absolute responses being derived from broadband illumination conditions.

Directional response measurements were taken for incident angles of the source aligned in two planes, both orthogonal to each other and to the surface of the spectral watch; the first plane passing through all three sensors simultaneously which uniquely determines the second plane for each sensor (Figure 1). The spectral watches were rotated around a vertical axis passing through the centre of the green sensor at intervals of 10°.

Linearity measurements were made using a range of broadband illumination conditions. The relevant descriptions of the illumination conditions depend on the spectral sensitivity results, so are stated later in Section 3.3. Absolute measurements of the spectral irradiance of the light incident on the sensors in these conditions, taking into account the known spectral sensitivity of the sensors already calculated, were used to calibrate the sensors.

3. Results

3.1. Spectral sensitivity

The spectral watches were exposed to a series of narrow-band illuminations between 360 and 720 nm, spaced at intervals of 6–13 nm, and with a bandwidth (full-width half-maximum) of 14–44 nm. The spectral sensitivities of the sensors in a typical spectral watch are shown in Figures 2 and 3 and given in Table 1.

'White light' readings are also produced. The 'white light' output is similar to green sensor data with small additions from the blue and red sensor data; so this is not reproduced in Figures 2 and 3.

Care was taken to ensure sufficient illumination for a linear response from each of the sensors. The high value of the variability for the green sensor response at 470 nm was mitigated by combining the results with the blue sensor results. This is discussed in more detail later.

3.2. Directional response

The rectangular window illustrated in Figure 1 measured approximately 10 mm × 3 mm and marked the extent of the transparent glass face, with the sensors set back around 2 mm from this surface. Parallax resulted from the sensors being set behind this window, which also created cylindrical asymmetry in the directional responses, illustrated in Figure 4, in the horizontal plane of incidence shown in the left panel of Figure 1. In other words, most lines of sight to the sensors more than $\pi/4$ radians off-axis are obstructed by the edge of the window, suggesting a substantial limitation in the directional response.

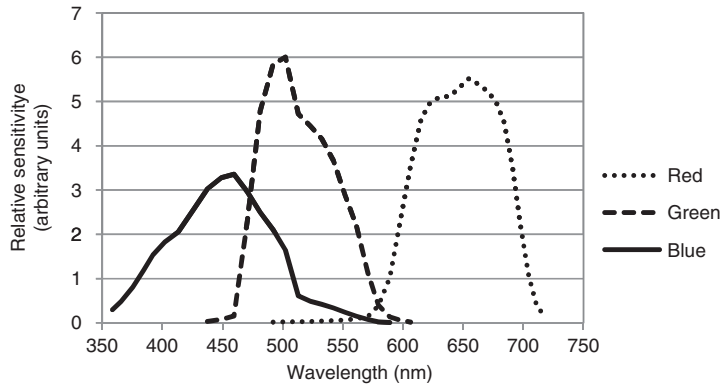


Figure 2 Mean relative spectral sensitivity of the sensors ($n = 16$ red and green $n = 15$ blue)

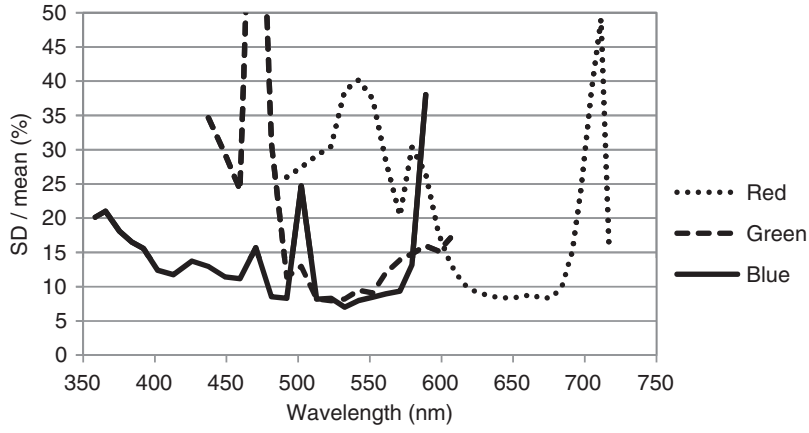


Figure 3 Variability in spectral sensitivity of the sensors between the spectral watches (SD over mean relative spectral sensitivity)

Table 1 Optical sensory performance of a typical spectral watch

Spectral response	Blue sensors	Green sensors	Red sensors
Peak response wavelength (nm)	460	500	655
Full-width half-maximum (nm)	395–500	475–550	600–695
Agreement better than 20% (nm) (Figure 3)	375–490	490–605	600–690

Figure 4 shows that in this horizontal plane, the response was closer to a cosine response, but aligned in the vertical plane (as illustrated in the right panel of Figure 1); there was less variability in the directional response between the sensors, as shown in Figure 5. This suggested that the horizontal positioning of the sensors relative to the lozenge-shaped window was more variable than its vertical positioning. The blue sensor

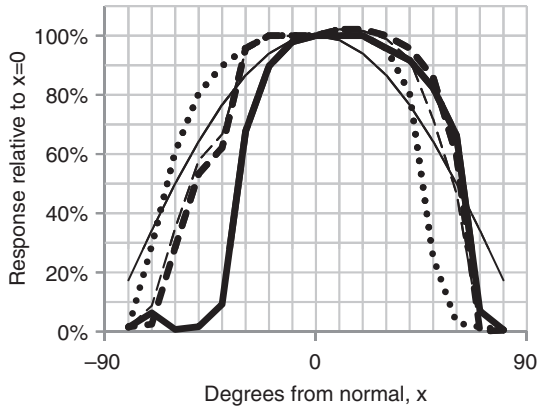


Figure 4 Example directional responses from each component of output for one spectral watch in the horizontal orientation. The thin black line shows the cosine response. The solid, dashed and dotted thick black lines show the blue, green and red sensor responses, respectively. The grey line shows the 'white light' output

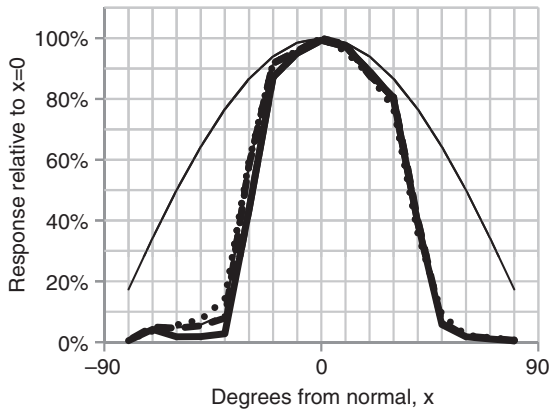


Figure 5 Example directional responses from each component of output for one spectral watch in the vertical orientation. The thin black line shows the cosine response. The solid, dashed and dotted thick black lines show the blue, green and red sensor responses, respectively. The grey line shows the 'white light' output

was the most compromised by its position on the spectral watch.

An analytic approach, using an adapted f_2 statistic for errors in directional response for weighted irradiance measurements ('cosine response errors'),[6] confirmed the more open horizontal geometry produced lower cosine response errors, but there was greater variability in cosine response errors between the spectral watches and between individual sensors in each (Table 2). The main contribution to the cosine response error came from angles outside $\pi/4$ radians due to parallax, as expected from an inspection of the sensor geometry.

3.3. Linearity and characterisation

The red, green and blue spectral watch outputs, as directly downloaded from the spectral watches, were stated in broadband units ($\mu W \cdot cm^{-2}$). These are clearly not radiometric units. However, the spectral sensitivity functions that defined these units were unspecified. It was unclear whether the spectral watches had shared units or, to some extent, unique units depending on variations between their spectral sensitivity functions.

To establish an unambiguous unit of measurement with the added benefit of a traceable calibration, the output data were recalibrated to create a common unit (recalibration factors for $\mu W \cdot cm^{-2}$ are shown in Figures 6 and 7) based on the mean sensitivity functions. The neutral density filters used reduced the transmission of broadband light to 49.78%, 10.82%, 1.24%, 0.35% and 0.27% of their original output weighted by the mean blue sensitivity function, and reduced the transmission to similar values with other weighting functions.

The output was also given in photon counts, which must necessarily be an approximation based on the broadband sensitivity nature of the sensors. With a common unit as described above, a documented conversion to approximate photon counts could be calculated.

The calibration factors given in Table 3 when multiplied with the spectral watch output provide measures of photometric quantities, normalised so that a source with

uniform spectral irradiance of $1\,\mu W\cdot cm^{-2}nm^{-1}$ should produce an answer of $1\,\mu W\cdot cm^{-2}$. In these same units, the weighted irradiances of the unattenuated source ($T = 2900\,K$) used to measure linearity were 104, 45 and $22\,\mu W\cdot cm^{-2}$ for red, green and blue sensors, respectively, and the lower bounds for linear response were found to be approximately 1% of these values (Figures 6 and 7).

Table 2 Mean percentage cosine response errors, $f_2 \pm$ SD between the spectral watches over negative and positive angles in the two planes of incidence, $n = 16$

Output	Plane of incidence	f_2 (%)	Plane of incidence	f_2 (%)
'White light' output	Horizontal	25.3 ± 1.0	Vertical	61.1 ± 1.2
Red sensors	Horizontal	33.2 ± 2.1	Vertical	60.9 ± 1.4
Green sensors	Horizontal	32.6 ± 1.5	Vertical	61.0 ± 1.4
Blue sensors	Horizontal	48.6 ± 4.7	Vertical	64.7 ± 1.7

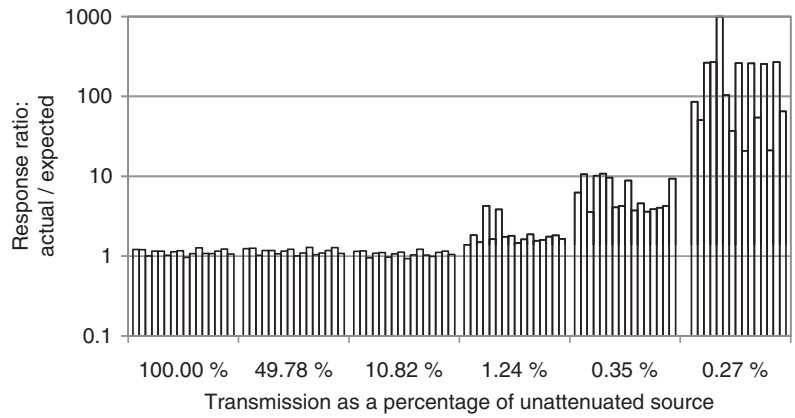


Figure 6 Relative calibration coefficients for all 16 blue sensors in broadband illumination

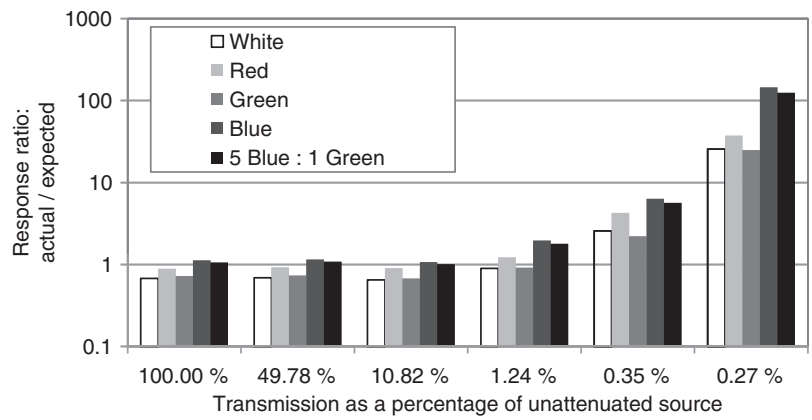


Figure 7 Mean calibration coefficients for all the sensors plus the synthesised output ('5 blue:1 green'), ($n = 16$, except $n = 15$ for 0.27% transmission)

4. Discussion

Both original data and subsequent detailed measurements of melatonin suppression and post-stimulus sustained pupilliary response show peak circadian response to wavelengths of around 480 nm, when log-fitted to a visual pigment template consistent with the spectral sensitivity characteristics of melanopsin.[11–15]

An appropriate linear combination of blue and green measurements can be used to mimic this action spectrum (Figure 8). Linear combinations between 10:1 and 2:1 (blue:green) reduce the high variability in the region 460–500 nm, as illustrated in Figure 3. However, the directional response and lower bound for linear response are not significantly improved for the blue sensor. The short-wavelength tail of the combined sensitivity remains higher than desired for an ideal match to the circadian/melanopsin action spectrum.

Linearly combining blue and green sensor outputs in a ratio of 5:1 produces a bell-shaped sensitivity curve with a peak sensitivity of around 480 nm (full-width half-maximum 400–510 nm, agreement better than 20% over 375–615 nm).

Peak circadian sensitivity for individuals with aged, yellowed lenses would be at marginally higher wavelengths.[16] Such circadian action spectra with peak sensitivity above 480 nm become harder to synthesise with the data from the spectral watches. Increasing the proportion of green beyond 4:1 is done at the cost of the overall shape of the curve and the gradual increase in the variability in the region of primary interest, 460–500 nm.

The use of diffusers to improve the directional response was investigated. Overall, incorporation of diffusers resulted in little improvement in directional responses, and increased the lower bound of the linear dynamic range. The clearest improvements in directional responses were for red sensors, whereas the blue sensors were least improved

Table 3 Average calibration factors and SD for red, green and blue sensors, $n = 16$

	Factor ± SD $\mu W \cdot cm^{-2}$/output unit
Red sensors	0.888 ± 0.059
Green sensors	0.722 ± 0.055
Blue sensors	1.122 ± 0.085

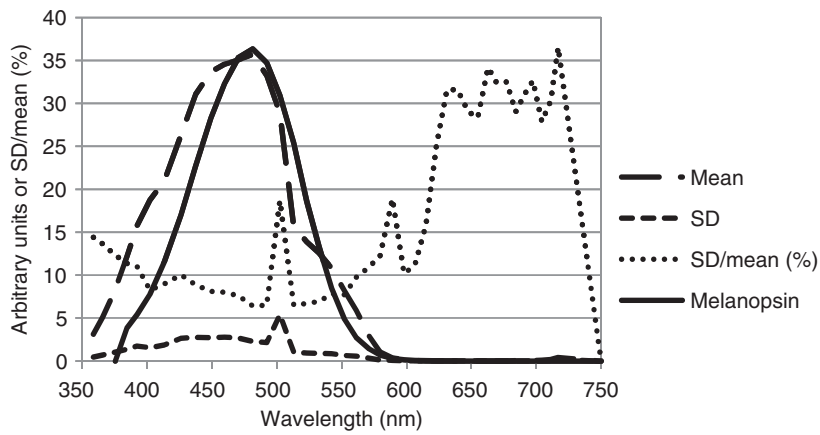


Figure 8 Mean spectral sensitivity and the associated SD of synthesised output for the circadian action spectrum, calculated in the ratio of 5 blue:1 green and the spectral sensitivity of melanopsin

and sometimes worsened in their directional response. The mutual proximity of the sensors makes it impractical to apply different diffusers externally for the three sensors.

The use of filters for approximating the circadian action spectrum also proves unnecessary as the synthesised spectral response illustrated in Figure 8 is a reasonable match for the current melanopsin-template based melatonin suppression action spectrum for blue light ($R^2 = 0.908$), which, the authors consider, represents the circadian action spectrum. Note that the peak response exceeds by around 30 nm that from the action spectrum produced in draft German pre-standards,[17] and has a reduced overlap with the blue light hazard action spectrum,[18] which should allow for less hazardous stimuli to be used.[19]

In addition to the performance factors discussed above, there are other relevant factors which may affect studies in which personal exposure meters are used. Two important factors that can be monitored are the time-keeping and change in response over the life-time of the personal exposure meter and, if applicable, its battery. The reduction in variability when combining output from the blue and green sensors from the spectral watches might suggest that the sensors compete for current. Other unconfirmed factors have suggested to the authors that the output may be reduced when the available power is lower. It would be useful to investigate the performance of the spectral watches with age, particularly the linearity of response to a known broadband source and the performance of the spectral watches in a variety of operating configurations. The data presented illustrate the output, whilst the spectral watches were collecting the maximum allowed data.

5. Conclusions

The spectral watches' sensors provided roughly linear output for weighted irradiance between around $1\,\mu W \cdot cm^{-2}$ to over $4000\,\mu W \cdot cm^{-2}$ in the spectrally weighted units described in Section 3.3, following spectral sensitivity curves with peak sensitivity and bandwidths 460 nm (395–500 nm), 500 nm (475–550 nm) and 655 nm (600–695 nm). The directional response was dominated by cylindrically asymmetric parallax and, hence, was not close to a perfect cosine response, with cosine response errors of approximately 30–50% in the horizontal plane and approximately 60–65% in the vertical plane.

The 'white light' output was close to a linear combination of the output of the three sensors. This output exhibited a superior directional response under the broadband tungsten-halogen illumination conditions. The 'white light' spectral sensitivity function had three separate peaks and the variability of the output between the spectral watches was high in the range 460–500 nm.

In contrast, a linear combination of the blue and green sensors output in the ratio 5:1 reduced the variability between the spectral watches in the range 460–500 nm, and provided a spectral sensitivity close to the circadian action spectrum.

This approach, alongside careful placement of the spectral watches, informed by their directional response characteristics, can be used to gather information on the circadian stimulus available to the eye at fixed locations in architectural spaces down to moderately low exposure conditions.

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